

DropBox

Functional requirements:

user should be able to upload/download any file based on his preference,
user should be able to search for any file among the existing files based on his preference,
user should be able to open/access any file based on his preference

Non-functional requirements:

data-partitioning,
load balancing,
fault tolerance,
vertical scalability,
horizontal scalability,
low latency,
high throughput,
CAP theorem (Here eventual consistency is fine but here we need 99.99% uptime i.e
Availability>>Consistency)

Core Entities:

user,
file,
filemetadata,
history

API:

POST:upload/file

```
{  
  userID,  
  {filemetadata}  
}
```

GET:/search/{fileID}

HLD:

